

Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
#1  
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
setwd("/home/guest/EDE_Fall2025/Data/Raw/Ozone_TimeSeries")  
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr   1.5.1  
## v ggplot2    4.0.0      v tibble    3.2.1  
## v lubridate  1.9.4      v tidyr     1.3.1  
## v purrr      1.1.0  
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()     masks stats::lag()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
install.packages("tidyverse")
```

```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'  
## (as 'lib' is unspecified)
```

```
library(lubridate)  
install.packages("lubridate")
```

```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'  
## (as 'lib' is unspecified)
```

```
library(zoo)
```

```
##  
## Attaching package: 'zoo'  
##  
## The following objects are masked from 'package:base':  
##  
##    as.Date, as.Date.numeric
```

```
install.packages("zoo")
```

```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'  
## (as 'lib' is unspecified)
```

```
install.packages("trend")
```

```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'  
## (as 'lib' is unspecified)
```

```
library(trend)  
list.files("/home/guest/EDE_Fall2025/Data/Raw/Ozone_TimeSeries")
```

```
## character(0)
```

```
install.packages("dplyr")
```

```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'  
## (as 'lib' is unspecified)
```

```
install.packages("purrr")
```

```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'  
## (as 'lib' is unspecified)
```

```
install.packages("readr")
```

```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'  
## (as 'lib' is unspecified)
```

```
library(dplyr)  
library(purrr)  
library(readr)  
install.packages("ggplot2")
```

```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'  
## (as 'lib' is unspecified)
```

```
library(ggplot2)  
install.packages("Kendall")
```

```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'  
## (as 'lib' is unspecified)
```

```
library(Kendall)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#2  
ozone2010 <- read_csv("EPAair_03_GaringerNC2010_raw.csv") %>%  
  mutate(Date = as.Date(Date, format = "%d/%m/%Y"))  
ozone2011 <- read_csv("EPAair_03_GaringerNC2011_raw.csv") %>%  
  mutate(Date = as.Date(Date, format = "%d/%m/%Y"))  
ozone2012 <- read_csv("EPAair_03_GaringerNC2012_raw.csv") %>%  
  mutate(Date = as.Date(Date, format = "%d/%m/%Y"))  
ozone2013 <- read_csv("EPAair_03_GaringerNC2013_raw.csv") %>%  
  mutate(Date = as.Date(Date, format = "%d/%m/%Y"))  
ozone2014 <- read_csv("EPAair_03_GaringerNC2014_raw.csv") %>%  
  mutate(Date = as.Date(Date, format = "%d/%m/%Y"))  
ozone2015 <- read_csv("EPAair_03_GaringerNC2015_raw.csv") %>%  
  mutate(Date = as.Date(Date, format = "%d/%m/%Y"))  
ozone2016 <- read_csv("EPAair_03_GaringerNC2016_raw.csv") %>%  
  mutate(Date = as.Date(Date, format = "%d/%m/%Y"))  
ozone2017 <- read_csv("EPAair_03_GaringerNC2017_raw.csv") %>%  
  mutate(Date = as.Date(Date, format = "%d/%m/%Y"))  
ozone2018 <- read_csv("EPAair_03_GaringerNC2018_raw.csv") %>%  
  mutate(Date = as.Date(Date, format = "%d/%m/%Y"))  
ozone2019 <- read_csv("EPAair_03_GaringerNC2019_raw.csv") %>%  
  mutate(Date = as.Date(Date, format = "%d/%m/%Y"))  
ozone2010 <- ozone2010 %>%  
  rename(  
    Date = `Date`,
```

```

    Daily.Max.8.hour.Ozone.Concentration =
      `Daily Max 8-hour Ozone Concentration`,
    DAILY_AQI_VALUE = `DAILY_AQI_VALUE`
  )
ozone2011 <- ozone2011 %>%
  rename(
    Date = `Date`,
    Daily.Max.8.hour.Ozone.Concentration =
      `Daily Max 8-hour Ozone Concentration`,
    DAILY_AQI_VALUE = `DAILY_AQI_VALUE`
  )
ozone2012 <- ozone2012 %>%
  rename(
    Date = `Date`,
    Daily.Max.8.hour.Ozone.Concentration =
      `Daily Max 8-hour Ozone Concentration`,
    DAILY_AQI_VALUE = `DAILY_AQI_VALUE`
  )
ozone2013 <- ozone2013 %>%
  rename(
    Date = `Date`,
    Daily.Max.8.hour.Ozone.Concentration =
      `Daily Max 8-hour Ozone Concentration`,
    DAILY_AQI_VALUE = `DAILY_AQI_VALUE`
  )
ozone2014 <- ozone2014 %>%
  rename(
    Date = `Date`,
    Daily.Max.8.hour.Ozone.Concentration =
      `Daily Max 8-hour Ozone Concentration`,
    DAILY_AQI_VALUE = `DAILY_AQI_VALUE`
  )
ozone2015 <- ozone2015 %>%
  rename(
    Date = `Date`,
    Daily.Max.8.hour.Ozone.Concentration =
      `Daily Max 8-hour Ozone Concentration`,
    DAILY_AQI_VALUE = `DAILY_AQI_VALUE`
  )
ozone2016 <- ozone2016 %>%
  rename(
    Date = `Date`,
    Daily.Max.8.hour.Ozone.Concentration =
      `Daily Max 8-hour Ozone Concentration`,
    DAILY_AQI_VALUE = `DAILY_AQI_VALUE`
  )
ozone2017 <- ozone2017 %>%
  rename(
    Date = `Date`,
    Daily.Max.8.hour.Ozone.Concentration =
      `Daily Max 8-hour Ozone Concentration`,
    DAILY_AQI_VALUE = `DAILY_AQI_VALUE`
  )

```

```

ozone2018 <- ozone2018 %>%
  rename(
    Date = `Date`,
    Daily.Max.8.hour.Ozone.Concentration =
      `Daily Max 8-hour Ozone Concentration`,
    DAILY_AQI_VALUE = `DAILY_AQI_VALUE`
  )
ozone2019 <- ozone2019 %>%
  rename(
    Date = `Date`,
    Daily.Max.8.hour.Ozone.Concentration =
      `Daily Max 8-hour Ozone Concentration`,
    DAILY_AQI_VALUE = `DAILY_AQI_VALUE`
  )
GaringerOzone <- bind_rows(
  ozone2010, ozone2011, ozone2012, ozone2013, ozone2014,
  ozone2015, ozone2016, ozone2017, ozone2018, ozone2019
)

dim(GaringerOzone)

```

```
## [1] 3589 20
```

```
colnames(GaringerOzone)
```

```

## [1] "Date"
## [2] "Source"
## [3] "Site ID"
## [4] "POC"
## [5] "Daily.Max.8.hour.Ozone.Concentration"
## [6] "UNITS"
## [7] "DAILY_AQI_VALUE"
## [8] "Site Name"
## [9] "DAILY_OBS_COUNT"
## [10] "PERCENT_COMPLETE"
## [11] "AQ5_PARAMETER_CODE"
## [12] "AQ5_PARAMETER_DESC"
## [13] "CBSA_CODE"
## [14] "CBSA_NAME"
## [15] "STATE_CODE"
## [16] "STATE"
## [17] "COUNTY_CODE"
## [18] "COUNTY"
## [19] "SITE_LATITUDE"
## [20] "SITE_LONGITUDE"

```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.

5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame `Days`. Rename the column name in `Days` to “Date”.
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame `GaringerOzone`.

```
# 3
class(GaringerOzone$Date)
```

```
## [1] "Date"
```

```
summary(GaringerOzone$Date)
```

```
##           Min.          1st Qu.          Median          Mean          3rd Qu.          Max.
## "2010-01-01" "2012-06-11" "2014-12-12" "2014-12-19" "2017-06-12" "2019-12-12"
##           NA's
##           "2168"
```

```
GaringerOzone <- GaringerOzone %>%
  mutate(Date = as.Date(Date, format = "%Y-%m-%d"))
# 4
GaringerOzone <- GaringerOzone %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)
# 5
Days <- data.frame(Date = seq(
  from = as.Date("2010-01-01"),
  to   = as.Date("2019-12-31"),
  by   = "day"
))
# 6
GaringerOzone <- Days %>%
  left_join(GaringerOzone, by = "Date")
```

Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

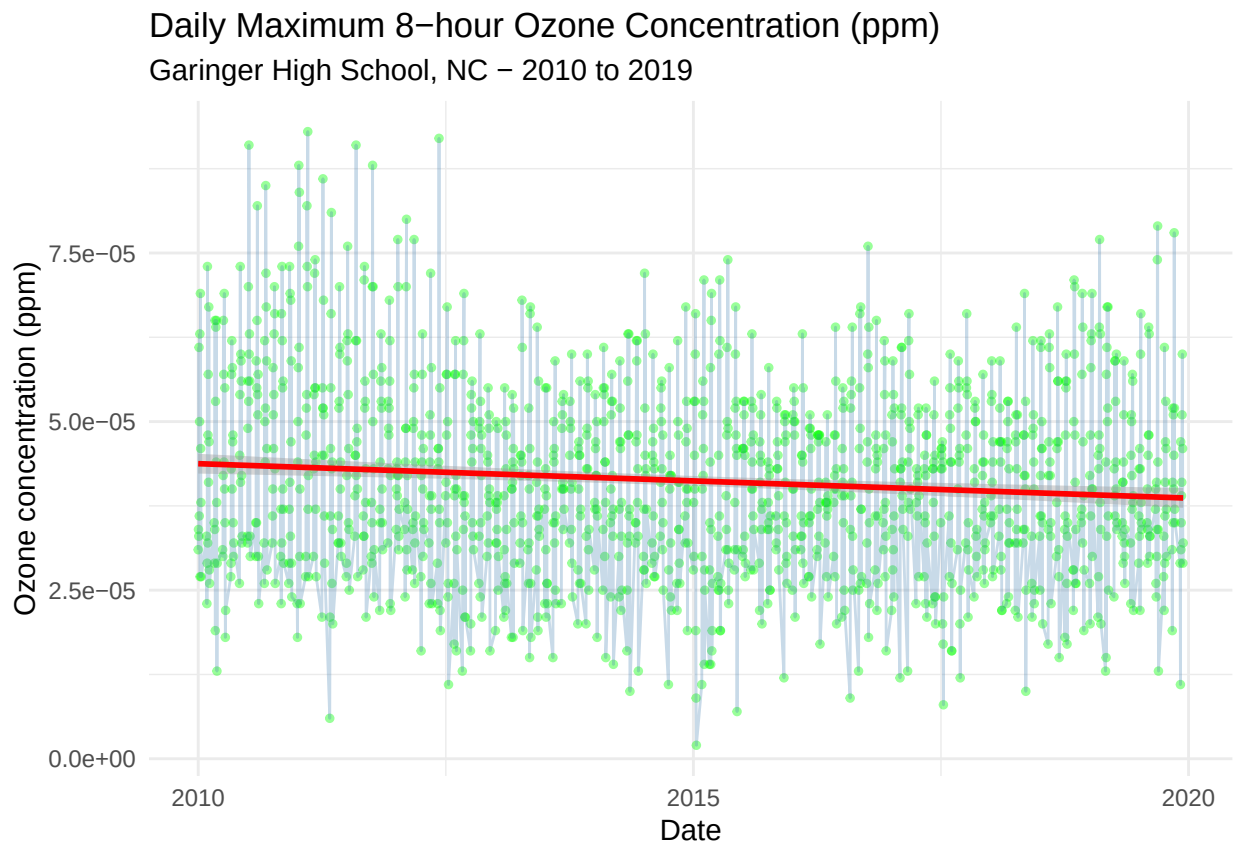
```
#7
GaringerOzone_plot <- GaringerOzone %>%
  mutate(ozone_ppm = Daily.Max.8.hour.Ozone.Concentration / 1000) %>%
  filter(!is.na(Date) & !is.na(ozone_ppm))
ggplot(GaringerOzone_plot, aes(x = Date, y = ozone_ppm)) +
  geom_point(alpha = 0.4, color = "green", size = 1) +
  geom_line(alpha = 0.3, color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "red", linewidth = 1) +
  labs(
```

```

title = "Daily Maximum 8-hour Ozone Concentration (ppm)",
subtitle = "Garinger High School, NC - 2010 to 2019",
x = "Date",
y = "Ozone concentration (ppm)"
) +
theme_minimal()

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Answer: The line plot for daily maximum 8-hour ozone concentrations at Garinger High School between 2010-2019 shows a downward-sloping linear trend. This indicates that ozone levels have generally decreased over this period. While there is noticeable seasonal variability, with higher ozone in summer and lower in winter, the overall trend suggests air quality has been increasing over time.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station? >Answer: Ozone concentrations at Garinger High School have decreased past 2010. The plot of daily maximum 8-hour ozone shows a downward-sloping linear trend, indicating a gradual decline. There are some seasonal variations but, overall the answer is yes.

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
GaringerOzone <- GaringerOzone %>%
  arrange(Date) %>%
  mutate(
    ozone_ppm = Daily.Max.8.hour.Ozone.Concentration / 1000,
    ozone_ppm_filled = na.approx(ozone_ppm, x = Date, na.rm = FALSE, rule = 2)
  )
```

Answer: A spline interpolation fits a smooth curve through all available points. While it may look nice, it can overshoot and produce unrealistic values. This may exaggerate trends or create false data ie. negative values. Piecewise constant fills any missing values with either the previous or next recorded value. This creates abnormalities in the data and given that ozone concentrations change daily, this would incorrectly display natural variability.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9
GaringerOzone.monthly <- GaringerOzone %>%
  mutate(
    year = year(Date),
    month = month(Date)
  ) %>%
  group_by(year, month) %>%
  summarise(
    ozone_ppm_mean = mean(ozone_ppm_filled, na.rm = TRUE),
    .groups = "drop"
  )
GaringerOzone.monthly <- GaringerOzone.monthly %>%
  mutate(
    Date = as.Date(paste(year, month, "01", sep = "-"))
  ) %>%
  select(Date, ozone_ppm_mean)
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

```
#10
GaringerOzone.daily.ts <- ts(
  data = GaringerOzone$ozone_ppm_filled,
  start = c(2010, 1),
  end = c(2019, 365),
  frequency = 365
)
GaringerOzone.monthly.ts <- ts(
  data = GaringerOzone.monthly$ozone_ppm_mean,
  start = c(2010, 1),
  end = c(2019, 12),
```

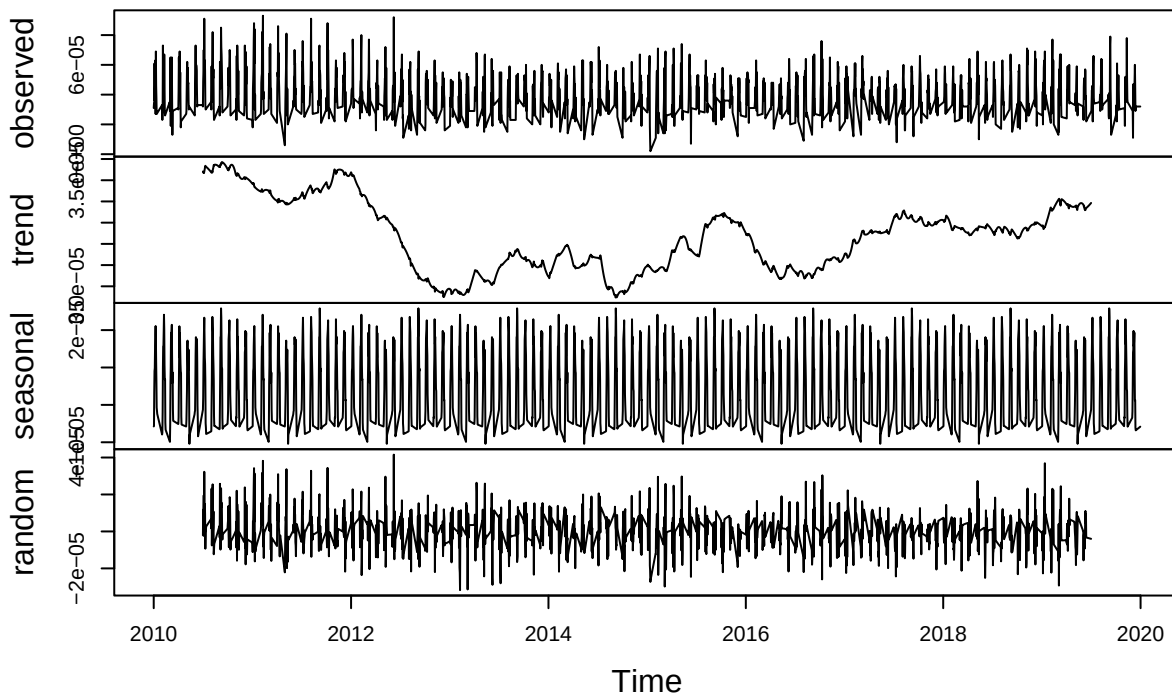


```
frequency = 12
)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

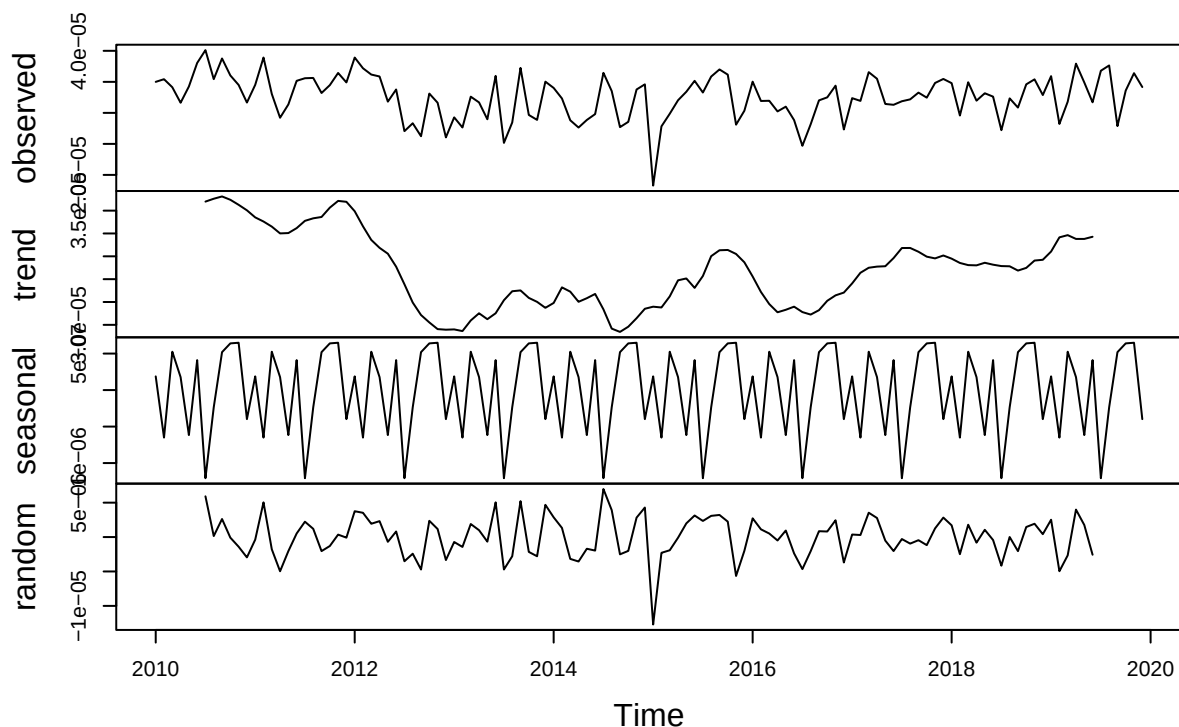
```
#11
GaringerOzone.daily.decomp <- decompose(GaringerOzone.daily.ts,
type = "additive")
plot(GaringerOzone.daily.decomp, xlab = "Time")
```

Decomposition of additive time series



```
GaringerOzone.monthly.decomp <- decompose(GaringerOzone.monthly.ts,
type = "additive")
plot(GaringerOzone.monthly.decomp, xlab = "Time")
```

Decomposition of additive time series



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12
ozone_monthly <- GaringerOzone.monthly$ozone_ppm_mean
mk_result <- SeasonalMannKendall(as.ts(GaringerOzone.monthly$ozone_ppm_mean))
print(mk_result)
```

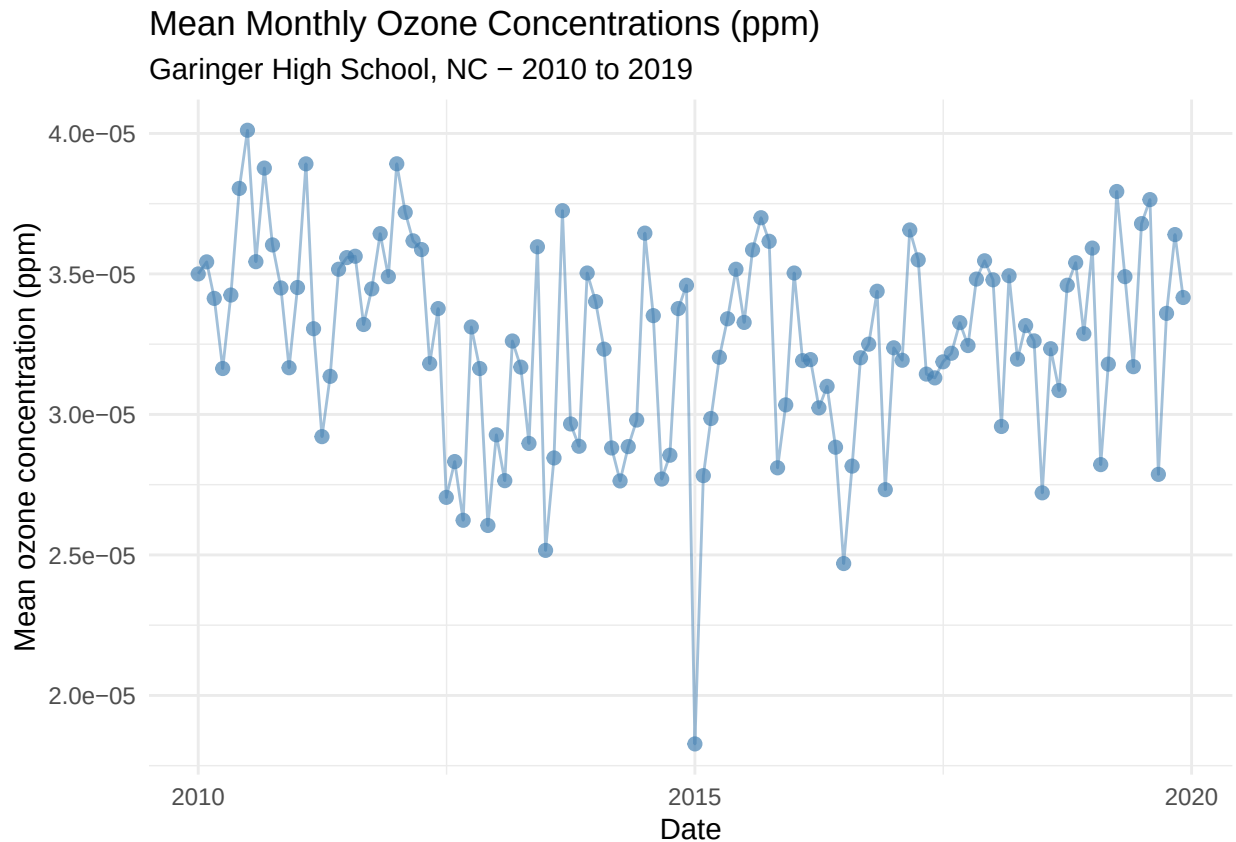
```
## tau = -0.058, 2-sided pvalue =0.3477
```

Answer: $\tau = -0.058$, 2-sided $p\text{-value} = 0.3477$. The seasonal Mann-Kendall test is appropriate for this data as it shows strong seasonality. This being true as ozone concentrations are higher in summer and lower in winter. A standard version of this test may mistake seasonal fluctuations as a trend. The seasonal version will in theory compare the same month across different years. Given the results, there was no strong evidence of a monotonic trend in monthly ozone concentrations. The negative τ suggests a small decline but the trend itself isn't meaningful.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
ggplot(GaringerOzone.monthly, aes(x = Date, y = ozone_ppm_mean)) +
  geom_point(color = "steelblue", size = 2, alpha = 0.7) +
  geom_line(color = "steelblue", alpha = 0.5) +
```

```
labs(
  title = "Mean Monthly Ozone Concentrations (ppm)",
  subtitle = "Garinger High School, NC - 2010 to 2019",
  x = "Date",
  y = "Mean ozone concentration (ppm)"
) +
theme_minimal()
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: There were seasonal patterns evident within the plot of mean monthly ozone concentrations at Garinger High School between 2010-2019. Higher ozone levels were present in the summer months and lower levels occurred in the winter. There also was a small downward trend in ozone concentration across the ten years which is not as noticeable as the seasonal fluctuations. Given the findings from the seasonal Mann-Kendall test, there was a small negative tau of -0.058 which is not statistically significant with a 2-sided p of 0.348. In sum, there is no strong evidence of a consistent long-term decline in monthly ozone concentrations apart from some seasonal variability.

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```

#15
GaringerOzone.monthly.decomp <- decompose(GaringerOzone.monthly.ts,
  type = "additive")
ozone_monthly_deseason <- GaringerOzone.monthly.ts -
  GaringerOzone.monthly.decomp$seasonal
#16
mk_deseason <- MannKendall(ozone_monthly_deseason)
print(mk_deseason)

```

```
## tau = -0.0671, 2-sided pvalue =0.27827
```

Answer: Following the removal of the seasonal component, the non seasonal Mann-Kendall test had $\tau = -0.067$ with a 2-sided p value of 0.278. By contrast, the seasonal Mann-Kendall results were $\tau = -0.058$ and p value of 0.348. Both tests indicated a small downward trend in ozone concentrations between 2010-2019. The test without seasons had a stronger negative τ and smaller p-value meaning that seasonal fluctuations had impacted the trend. It is important to note that the trend was not statistically significant with either test so there was not strong evidence of an overall consistent long term decline in monthly ozone.